

Final Project

Kutner C.7: Real Estate Sales

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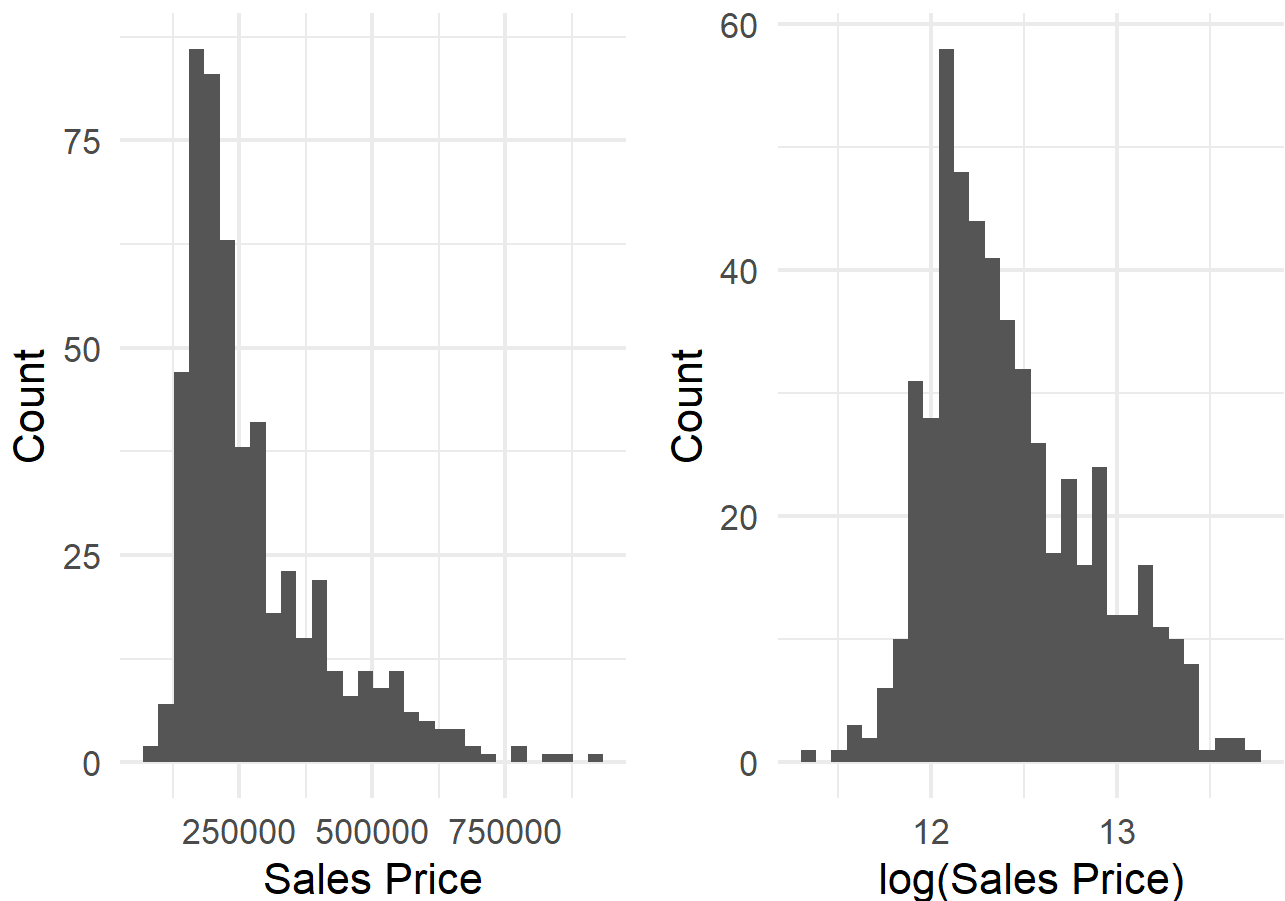
Dimensions of dataset: 522 13

Variable	Description/Type	Variable	Description/Type
ID	Data identification number; continuous predictor	Pool	Does the residence have a swimming pool; categorical predictor (0, 1 binary)
Sqft	Finished square footage of the property; continuous predictor	Year	Year the property was originally constructed; continuous predictor
Bed	Total number of bedrooms in the residence; continuous predictor	Quality	Index for quality of construction; categorical predictor (1=high quality, 2=medium quality, 3=low quality)
Bath	Total number of bathrooms in the residence; continuous predictor	Style	Qualitative indicator of architectural style; categorical predictor (1,2,3,4,5,6,7,9,10,11)
AC	Does the residence have AC; categorical predictor (0, 1 binary)	Lot	Size of the lot the residence is on; continuous predictor
Garage	Number of cars the garage will hold; continuous predictor	Highway	Is the residence adjacent to a highway; categorical predictor (0, 1 binary)
Price	Sales price of the residence in dollars; continuous response		

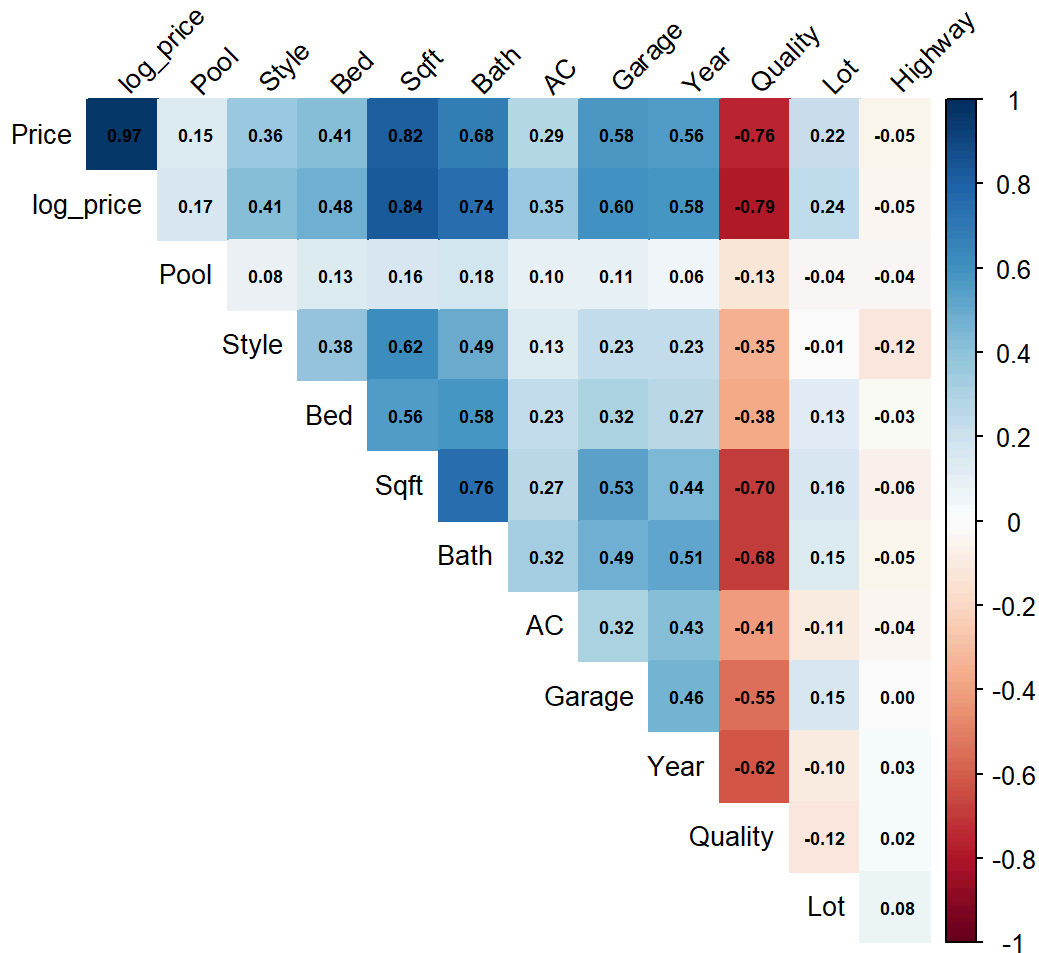
A few things to note about the data. As we can see there are 7 continuous predictors and 5 categorical predictors. The response variable is the sales price of the specific residence at the ID number, which is a

continuous value, so I am choosing to use a linear model considering the mix of predictors and the fact that the response is continuous. In all of the cases of a binary categorical predictor 0 represents "no" and 1 represents "yes". For the style parameter, there are 11 different categories. I was unable to find any sort of information on what the 11 styles represent so they will have to remain as arbitrary categories. The main design choice that I made is to represent the year, bath, bed, and garage predictors as continuous variables instead of categorical. Requiring dummy variables for each of these would create too many predictors. Also, for each of these values there is a genuine meaning for an increase, so the coefficients will have a good interpretation. Finally, ID does not actually represent anything so it will be excluded from all diagnostics and modeling. This leaves us with 6 continuous predictors and 5 categorical predictors, giving us 11 predictors to select from.

Based on these predictors, the proposed research question is: What combination of structural, quality, and location characteristics best predicts a residence's sale price, and how much of the variation can these features explain?

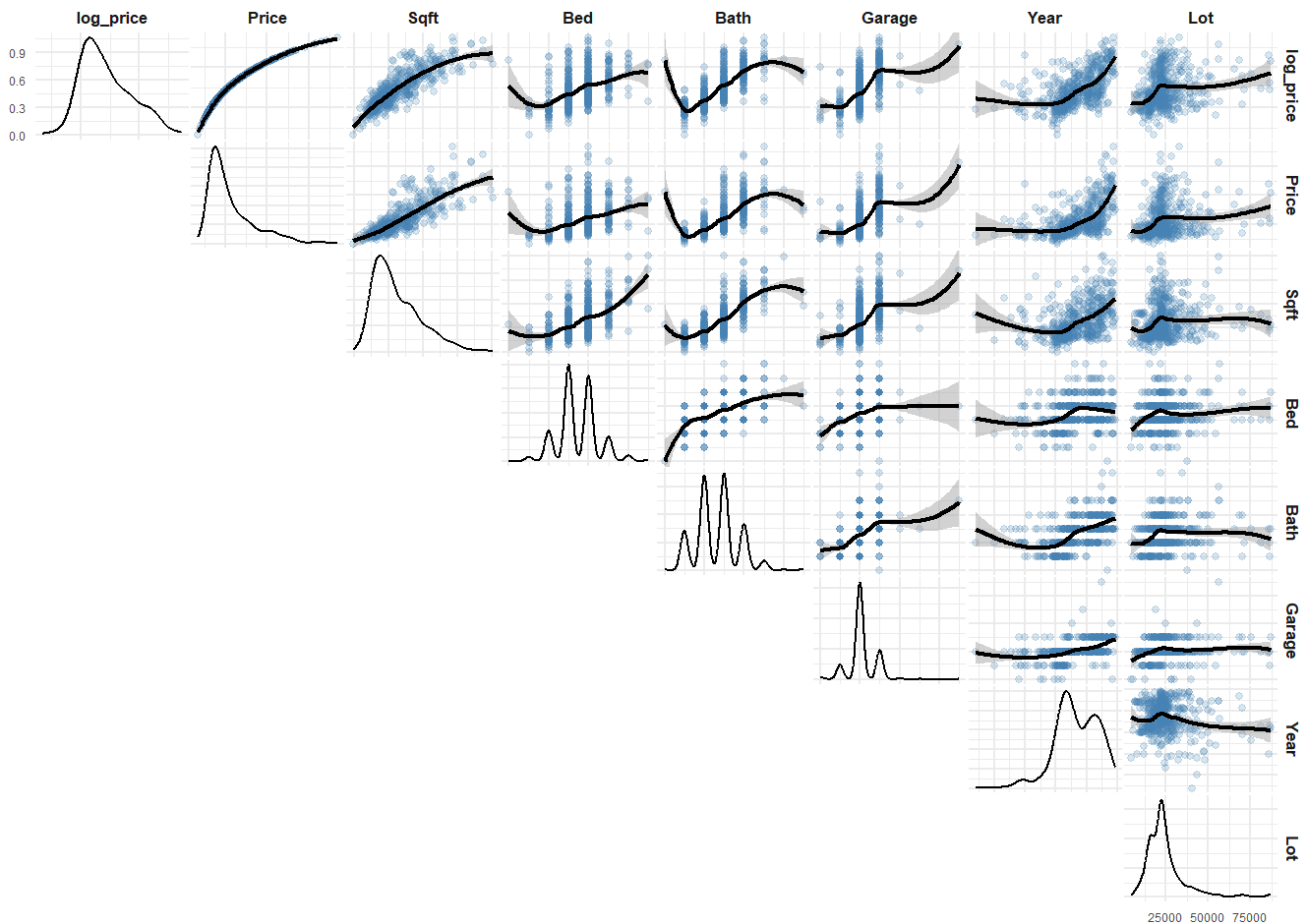


Since the response variable is sales price, I decided to plot the histogram along with the histogram of the log since price often grow exponentially and using log will smooth out some of the relationships. It is clear that the histogram of the price is very much right skewed which does make sense in the context of sales price of a residence. With the transform to $\log(\text{price})$, the histogram is still right skewed, but it is much less severe than without the transformation. Since we don't need symmetry in the response distribution this is not enough to fully commit to $\log_{\text{price}}$ and we will consider this question more as we explore further.

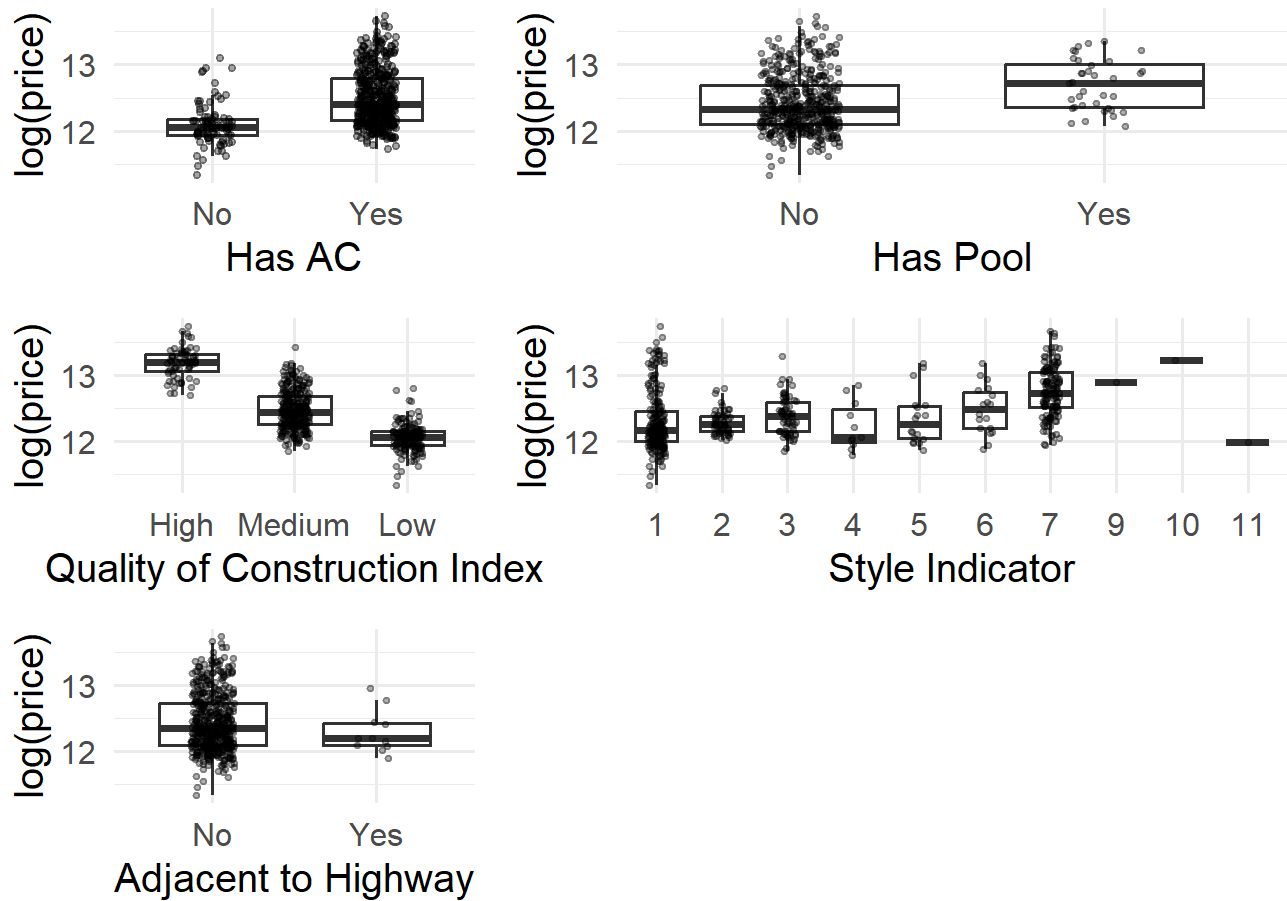


The first thing to analyze from the above plot is which predictors appear correlated with the response (which we have changed to log_price). Obviously price and log_price are highly correlated but we won't consider that because we will only use one of these response variables. Additionally, it is clear that log_price has a narrowly higher correlation with each predictor than price. This is still not enough evidence to commit to log_price, but for analysis of correlation we will use the values that correspond to the relationships of log_price, not price.

The highest correlation is between log_price and square feet with $\text{cor} = 0.84$. Some other predictors with high correlation are quality (-0.79), number of bathrooms (0.74), and number of cars that can fit in the garage (0.60). The two predictors with the strongest correlation are number of bathrooms and square feet, with $\text{cor} = 0.76$. There are several other pairs that are highly correlated such as square feet and quality (-0.70), number of bathrooms and quality (-0.68), year and quality (-0.62), and finally style and square feet (0.62). All of these predictor pairs with correlation magnitude over 0.6 are things to focus on for best subset selection and potential interaction terms.



The next step is to analyze the relationships between the continuous predictors and the response to see if nonlinear relationships exist. The curves for bed, bath, and garage (beyond!!) appear to contain some nonlinear relationship but for small and large values of each predictor the data is very sparse so it is hard to say if those nonlinear relationships are genuine or just due to sparsity. For lot, there is definitely a nonlinear curve at small values and then the curve flattens out into essentially a horizontal line. The strongest evidence for a nonlinear relationship is in the year predictor. There is a clear increasing curve for increasing year values, and that is where most of the data is. There is definitely potential to use a quadratic term for year to fit this data.



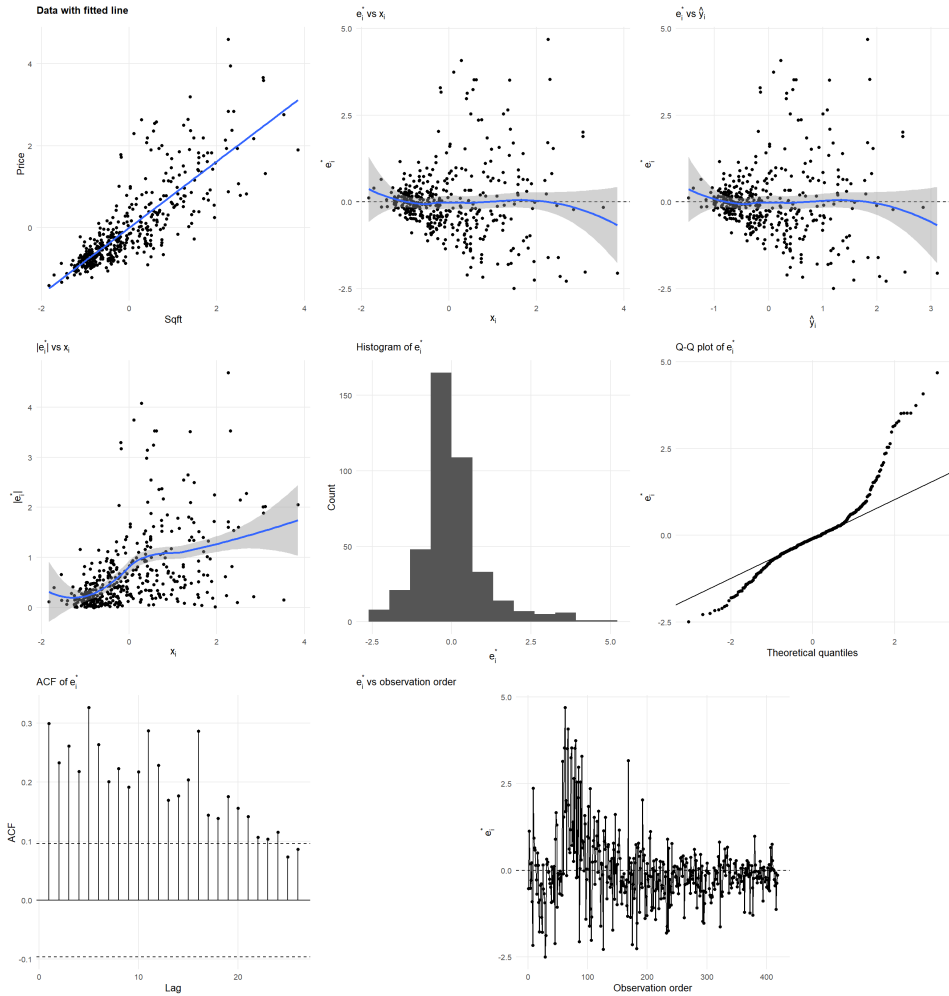
I chose to do `log_price` as the response for these plots even though I have not selected it as the response yet. It is currently the leading candidate and the relationships are the same regardless of whether I transformed the response or not.

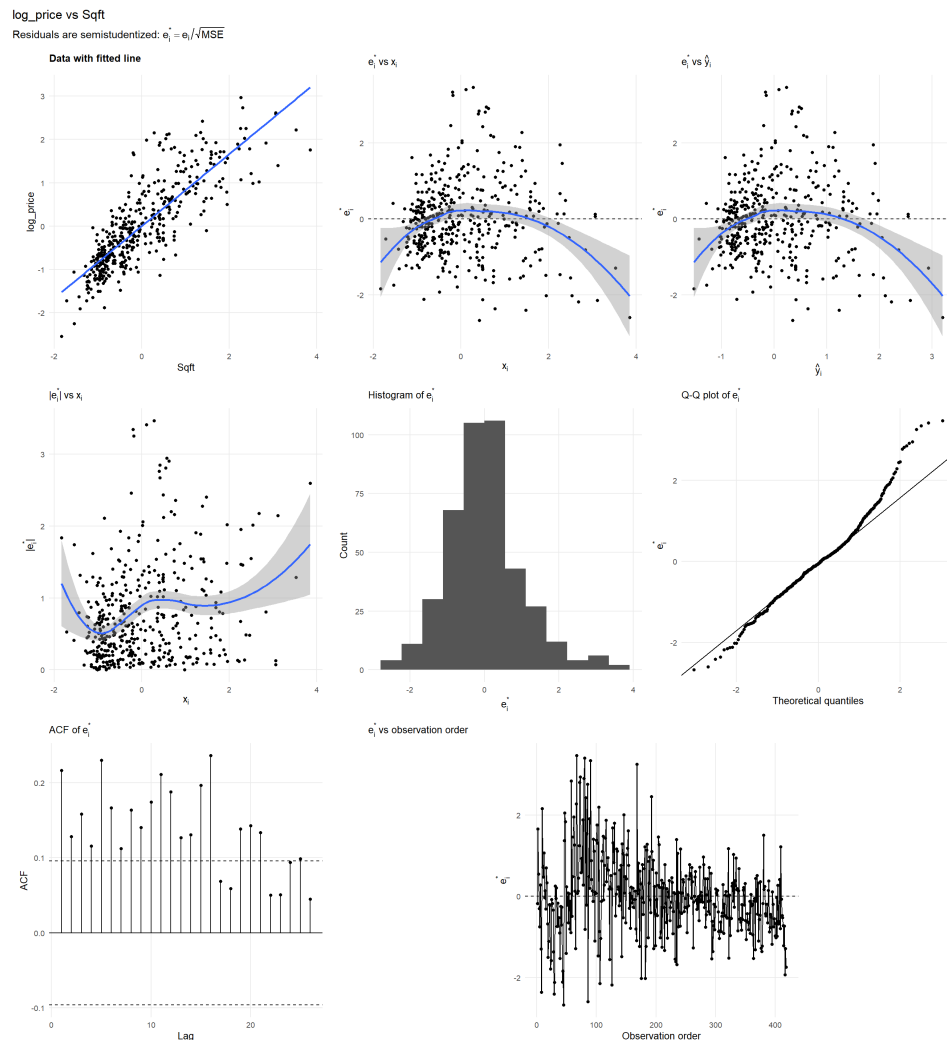
Now to do the same analysis for the categorical predictors. AC, pool, and style all show a pretty horizontal fit, there is not much difference from one group to another. In the style plot there is some variation for values 9, 10, and 11, but there is also little to no data there. There is a slightly increasing trend for both AC and pool but it is minimal. Highway similarly shows little to no variation from one group to the other. There is a strong linear relationship in the quality plot. Clearly `log_price` is negatively correlated with the construction quality, which makes sense given that 1 indicates high quality while 3 indicates low. Overall there doesn't appear to be any nonlinear relationship between any of the categorical predictors and `log_price`. At this point style is an extremely arbitrary parameter and the data is very sparse for all levels besides 1, 2, and 3. Because we can't interpret it anyways and it has weak predictive power overall based on the correlation heat map, I am choosing to omit it from this point on as it will only make the model more confusing without providing much benefit.

Standardized continuous predictors and split into training and test sets (80/20) before fitting models

training	test
418	104

Price vs Sqft
Residuals are semistudentized: $e_i^* = e_i / \sqrt{MSE}$





- Note that since this is not a time series, the ACF plot does not provide information as useful as our other diagnostic plots *

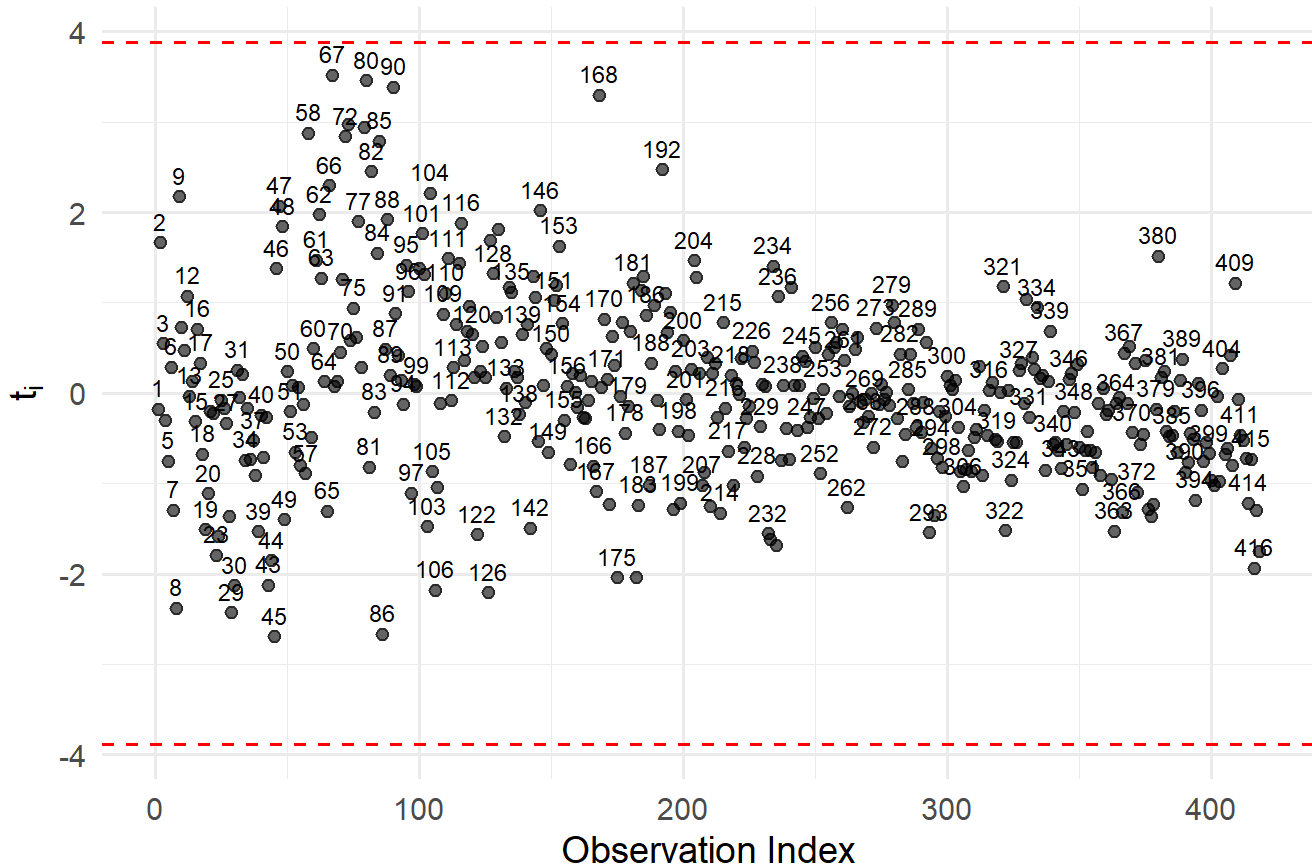
Here I plotted diagnostic for two SLR models using Sqft as the predictor since it had the highest linear correlation with the response. First I plotted Price vs Sqft, then transformed to log_price vs Sqft. Looking at the first set of plots with the untransformed response there is a very clear story. Looking at the plot of Price vs Sqft we can see that the data is fairly linear, but it blows up as Sqft increases creating a clear funnel shape. Looking at the residual plots we see that the confidence interval encompasses the zero line very well, indicating that there is in fact a linear relationship here. However, the funnel shape in these plots is obvious and severe. This combined with the shape of the first plot indicate heteroskedasticity. This is immediately confirmed in the absolute residual plot, which has a clear curve and funnel shape, confirming the suspicion of non constant variance. The histogram of residuals looks very slightly right skewed, while the QQ plot has noticeable heavy tails. This is likely due to the non constant variance issue we already observed.

Transforming the response to log_price does exactly what we expect. The log_price vs Sqft plot is extremely linear, and the blow up for larger values of Sqft is very minimal. Looking at the residual plots, the obvious funnel shape has disappeared, and while there is still some evidence of a diamond shape that would indicate heteroskedasticity it is not as extreme. This is confirmed in the absolute residual plot, which still has clear curves, but the absolute residuals are much more evenly scattered, and besides the dip for low values of Sqft, we can almost draw a horizontal line through the entire confidence band. The histogram is very

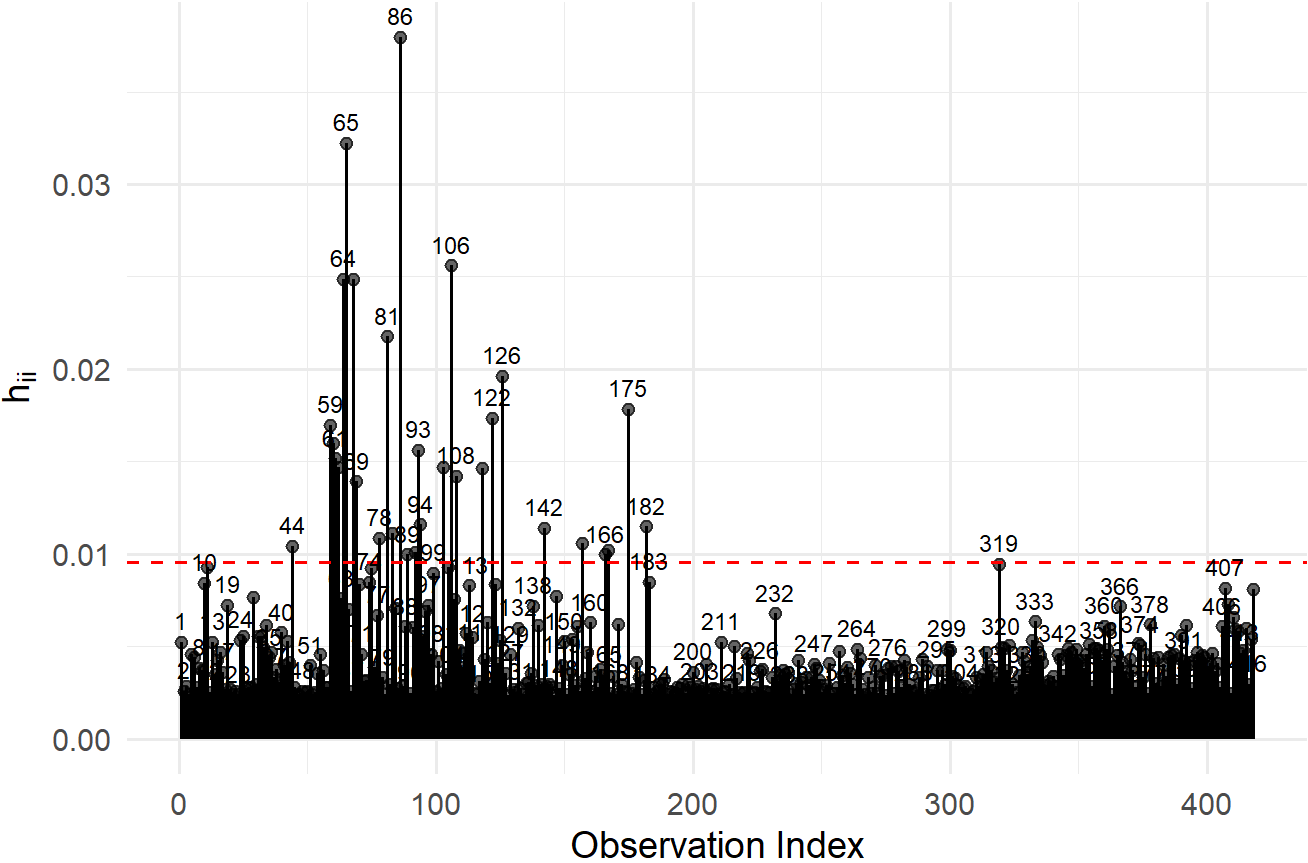
symmetric, and the QQ-plot still shows heavy tails but they are less drastic than before the transformation. On account of the symmetric histogram, these tails are certainly caused by the non constant variance observed in the absolute residual plot. Going back to the residual plots, there is a very obvious U-shape that indicates a possible relationship with $Sqft^2$ as a quadratic term. If I were to pursue this model further I would choose to fit it with weighted least squares in order to solve the non constant variance issue.

Overall, the log transform mitigates the issue of heteroskedasticity significantly so I will select it as the response variable going forward and throw out Price.

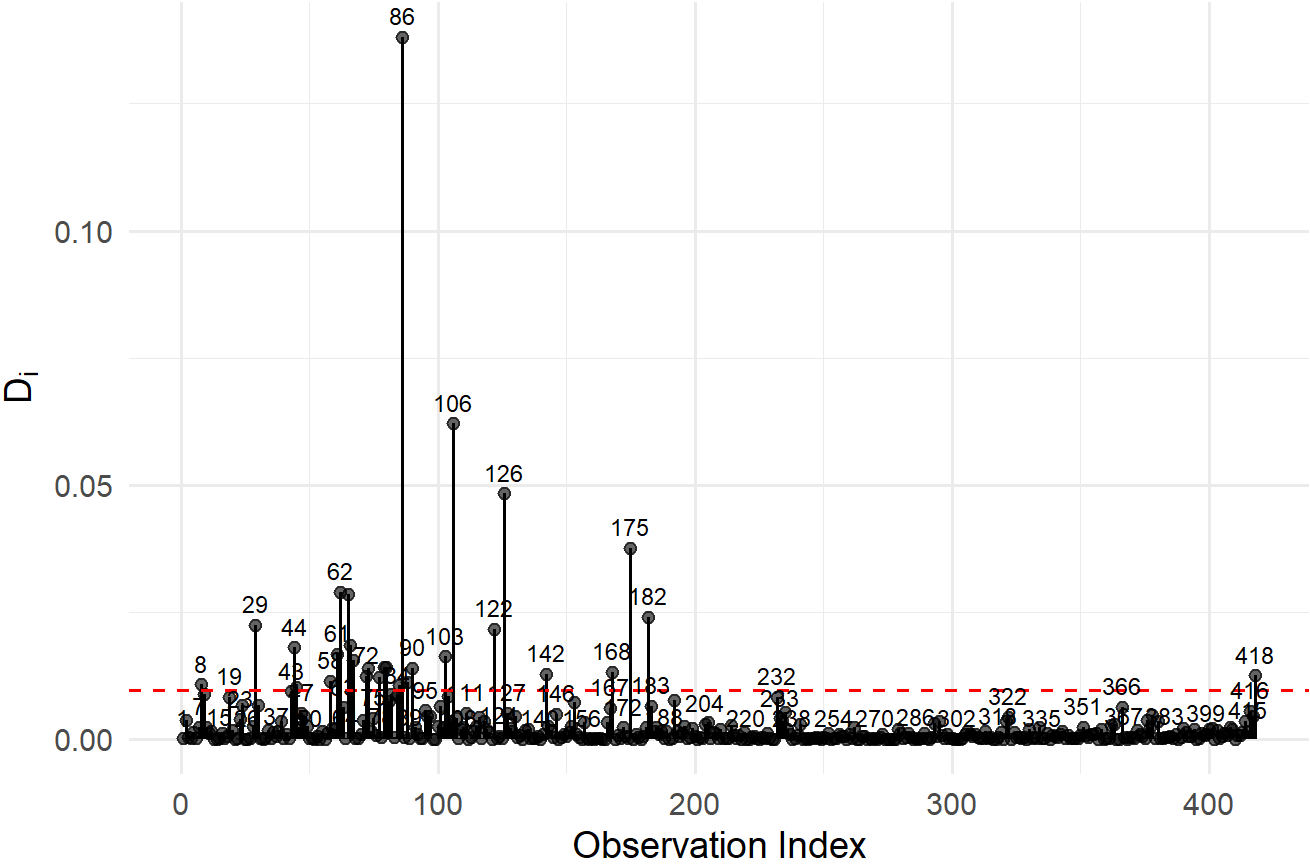
Externally Studentized Residuals



Leverage (Hat Values)



Cook's Distance



Looking at our outlier diagnostics, we can first see that none of the observations have ESR outside of our expected range, which is a good sign. This trend does not continue for the leverage as we can see several values above the threshold, with 86 and 65 being the highest by far. Then when we see our Cook’s distance plot, observations 86 and 106 have the most influence which makes sense considering the high leverage and the somewhat high ESR. The tables below show these things more quantitatively.

obs	ext_stud
67	3.516
80	3.456
90	3.387
168	3.295
73	2.976
79	2.937
58	2.875
72	2.836
85	2.788
45	-2.694

obs	leverage
86	0.038
65	0.032
106	0.026
68	0.025
64	0.025
81	0.022
126	0.020
175	0.018
122	0.017
59	0.017

obs	cooks_d
86	0.138
106	0.062
126	0.048
175	0.038
62	0.029
65	0.028
182	0.024
29	0.022
122	0.022
66	0.018

coefficient	full_data	without_observation	change
(Intercept)	0.000	0.004	0.004
Sqft	0.832	0.846	0.014

The highest influence observation was 86, so here I did a sensitivity analysis and removed it. After refitting the model without this point we can see that there is very minimal change in the model. The change is less than 1 percent so I don't think there is any reason to remove any of these points as they are not particularly influential. This confirms the earlier visual suspicion.

```

      fit      lwr      upr
1 300795.0 293138.2 308651.8
2 430758.3 413205.9 449056.3
3 515484.0 489556.4 542784.7

```

```

      fit      lwr      upr
1 300795.0 187601.9 482285.2

```

```

2 430758.3 268355.6 691443.4
3 515484.0 320822.4 828257.9

```

Here are predicted sales prices at 0.5, 1.5, and 2 standard deviations above the mean of log_price using confidence and prediction intervals. The first 3 fit columns gives us the mean predicted price at each respective value above the mean. As we can see there is a positive correlation between increased sqft and the resulting price, which is exactly what we expect. The second three values show us a single predicted price so it makes sense that the intervals are much wider.

Full predictor pool:

```

[1] "Sqft"    "Bed"      "Bath"      "AC"        "Garage"    "Pool"      "Year"
[8] "Quality" "Lot"      "Highway"

```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.578	0.095	6.062	0.000
Sqft	0.435	0.036	11.913	0.000
Bed	0.001	0.026	0.027	0.978
Bath	0.053	0.036	1.463	0.144
AC	0.141	0.063	2.238	0.026
Garage	0.061	0.025	2.446	0.015
Pool	0.172	0.077	2.223	0.027
Year	0.174	0.027	6.471	0.000
Quality2	-0.721	0.077	-9.317	0.000
Quality3	-0.973	0.107	-9.104	0.000
Lot	0.169	0.021	7.891	0.000
Highway	-0.185	0.125	-1.480	0.140

	GVIF	Df	GVIF^(1/(2*Df))
Sqft	3.42	1	1.85
Bed	1.71	1	1.31
Bath	3.32	1	1.82
AC	1.39	1	1.18
Garage	1.59	1	1.26
Pool	1.05	1	1.03
Year	1.86	1	1.36
Quality	3.24	2	1.34
Lot	1.17	1	1.08
Highway	1.03	1	1.01

Based on the results here we can analyze some things about this predictor pool. As far as significance of coefficients we can see that bed, bath, and highway (beyond again!!) are all insignificant while the rest of the predictors are significant. The highest VIF values are sqft, quality, and bath, but none of them are above 5 which is the stricter of the rules of thumb for VIF so multicollinearity doesn't seem like much of an issue here. All of the other predictors have very low VIF less than 2.

Original predictor pool size vs full predictor pool size:

```

original_preds    full_preds
          10              71

```

After adding all quadratic predictors and interaction terms this is the size of our predictor set. We can verify this result mathematically. Recall that there are 6 continuous predictors and 4 categorical predictors, 3 of which are binary and 1 which is a three level factor. The factor gets turned into 2 dummy variables, meaning that our original predictor pool is 11 predictors, 6 of which are continuous and 5 of which are binary. We square the 6 continuous predictors giving us a pool of 17. Finally, there are 11 choose 2 ways to make the interaction terms which comes out to 55 total terms. 11 original predictors plus 6 quadratic terms plus 55 total terms which gives us 72 total terms. However we have the interaction term between the two quality dummy variables which will always be zero, effectively giving us 71 terms in the full predictor pool.

Forward selection candidate model:

```
log_price ~ Lot + Pool + Quality2 + Quality3 + Sqft + Year +
  Bath2 + Bed2 + Year2 + `AC:Pool` + `Bath:Quality2` + `Sqft:Bed` +
  `Sqft:Quality2` + `Year:AC` + `Year:Quality2`
```

Backward selection candidate model:

```
log_price ~ AC + Lot + Pool + Year + Bath2 + Year2 + `AC:Pool` +
  `AC:Quality2` + `AC:Quality3` + `Bath:Quality2` + `Sqft:Quality2` +
  `Sqft:Quality3` + `Sqft:Year` + `Year:Quality3`
```

Here we have the formulas for the two candidate models, the forward selection model and backward selection model. Before any analysis we must consider the model fitting hierarchy. In the forward selection model, we have bath² and bed² so we would add the corresponding main effect terms. From the interaction terms we would only have to add AC as a main effect term. The model originally contained 15 predictors and adding these 3 gives us a model with 18 predictors (p = 19). For the backward selection model, from the quadratic terms we would have to add the main effect of bath. From the interaction terms we add quality2, quality3, and sqft. The original model had 14 terms and we added 4 more giving us 18 predictors once again (p = 19). In general, backward selection will produce a larger model but in this case the original model was actually smaller than forward selection, and the two models ended up being the same size, which is interesting to note. The structure of these formulas makes some sense intuitively. Since the backward selection process starts with all predictors present, it can freely drop main effects in favor of interaction terms, while the forward selection process has to build the model up from the main effects. This is why we see more main effects in the original formula of the forward model, and the backward model contains mostly interaction terms. This is what the final two candidate models look like after adding the necessary main effects. Additionally I have chosen to remove AC:Quality3 from the plot.

Number of residencies with a pool but no AC: 1

Observation where singularity occurs: 8

Leverage at this observation: 1

Before we move onto diagnostics, we need to examine a few of these interaction terms. The first term we want to look at is AC:Pool. As we can see from the output above there is only one observation such that AC

= 0 and Pool = 1. This creates a singularity in the data, which means that this observation will have a leverage of 1 (confirmed above). Leverage of 1 will give us a PRESS value and a Cook's distance of INF at this particular observation, which will make our model difficult to diagnose properly. Because of this we will remove it from the predictor set of both candidate models.

Forward selection candidate model:

```
log_price ~ AC + Bath + Bed + Lot + Pool + Quality2 + Quality3 +
  Sqft + Year + Bath2 + Bed2 + Year2 + `Bath:Quality2` + `Sqft:Bed` +
  `Sqft:Quality2` + `Year:AC` + `Year:Quality2`
```

Backward selection candidate model:

```
log_price ~ AC + Bath + Lot + Pool + Quality2 + Quality3 + Sqft +
  Year + Bath2 + Year2 + `AC:Quality2` + `AC:Quality3` + `Bath:Quality2` +
  `Sqft:Quality2` + `Sqft:Quality3` + `Sqft:Year` + `Year:Quality3`
```

Model :

```
log_price ~ AC + Pool + Year + Lot + Bath2 + Year2 + `Sqft:Year` +
  `Sqft:Quality2` + `Sqft:Quality3` + `Bath:Quality2` + `Year:Quality3` +
  `AC:Quality2` + `AC:Quality3` + Bath + Quality2 + Quality3 +
  Sqft
```

Complete :

	(Intercept)	AC	Pool	Year	Lot	Bath2	Year2	`Sqft:Year`	`Sqft:Quality2`
Quality3	1	-1	0	0	0	0	0	0	0
Quality3	0		0			0		1	
Quality3	1		0	-1		0			

The above output shows us that with the current equation for this model creates a design matrix that is not full rank. As we can see we can write Quality3 as $Quality3 = \text{Intercept} - AC + AC:Quality2 + AC:Quality3 - Quality2$, and this function is completely deterministic, so our design matrix columns are linearly independent. This creates a data leak in our model that will cause issues in fitting coefficient values and other diagnostics. Because of this I decided to remove one of the predictors in this dependency. I am going to remove AC:Quality3 because it is the least varied; most of the time when Quality3 is 1, AC is 0. The equations below are the final model equations from the full predictor set that I will be diagnosing.

Forward selection candidate model:

```
log_price ~ AC + Bath + Bed + Lot + Pool + Quality2 + Quality3 +
  Sqft + Year + Bath2 + Bed2 + Year2 + `Bath:Quality2` + `Sqft:Bed` +
  `Sqft:Quality2` + `Year:AC` + `Year:Quality2`
```

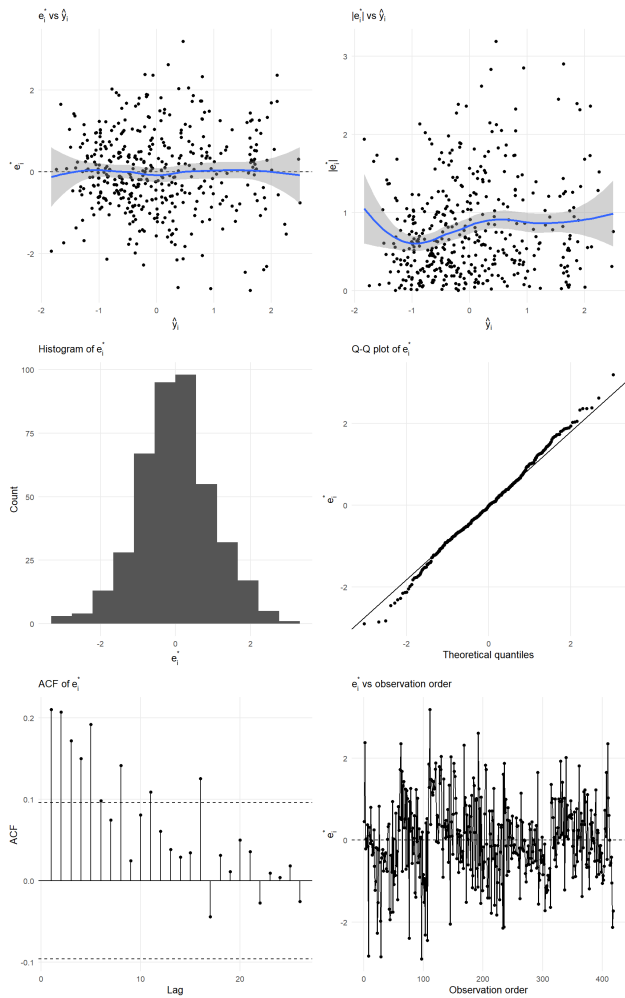
Backward selection candidate model:

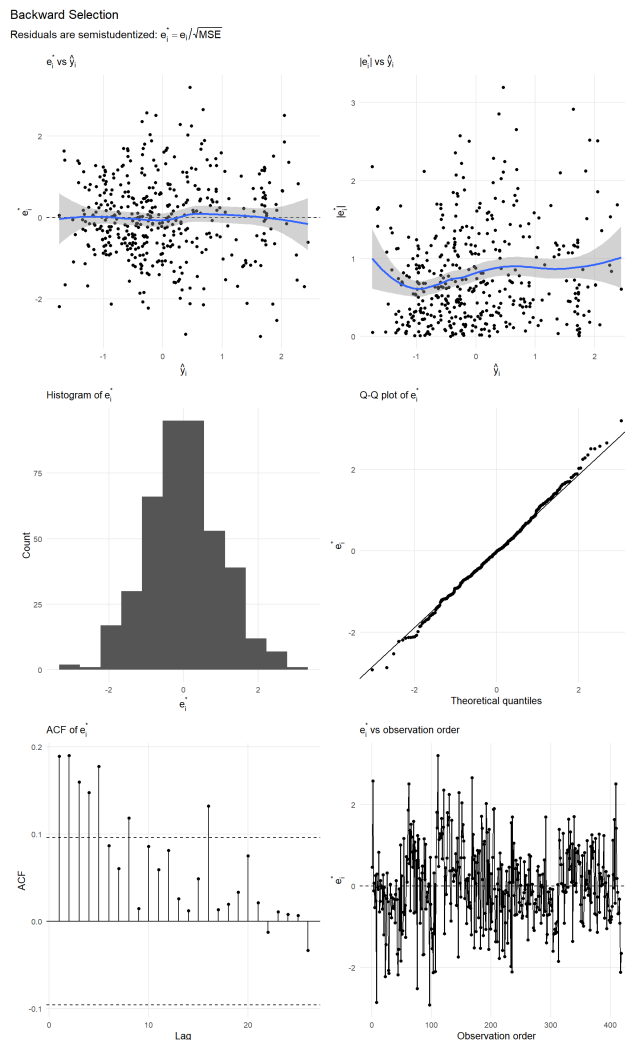
```
log_price ~ AC + Bath + Lot + Pool + Quality2 + Quality3 + Sqft +  
  Year + Bath2 + Year2 + `AC:Quality2` + `Bath:Quality2` +  
  `Sqft:Quality2` + `Sqft:Quality3` + `Sqft:Year` + `Year:Quality3`
```

Model	Predictors	Adj_R2	BIC	AIC	PRESS_RMSE	Test_RMSE
Full Main Effects Model	11	0.8377	492.5066	440.0454	0.4133	0.4372
Forward Selection	17	0.8654	444.3704	367.6963	0.3817	0.3954
Backward Selection	16	0.8652	439.7114	367.0728	0.3793	0.4130

Here is a comparison of the forward selection, backward selection, and a model with only our main effects. As we can see, the main effects model performs worse than the other two on all metrics. It performs worse on adjusted R^2 by a modest 4 percent, but then is significantly worse on AIC and BIC by around 20 percent and 10 percent respectively. The main effects model also performs worse on PRESS RMSE by about 7 percent. This aligns with our SLR diagnostics, which showed that a model with only linear main effects was likely insufficient to capture relationships. From this point on we will throw out the main effects model and diagnose the forward and backward selections. One thing to note at this point is that the forward and backward selection were done with BIC penalty. This choice was made to stay true to the research question. BIC penalty generally promotes a smaller model as it penalizes complexity more than AIC, giving us a smaller and more interpretable model. In this case it is important that we get a good understanding of what predictors have an impact on sales price so interpretability is very important.

Forward Selection
Residuals are semistudentized: $e_i^* = e_i / \sqrt{MSE}$

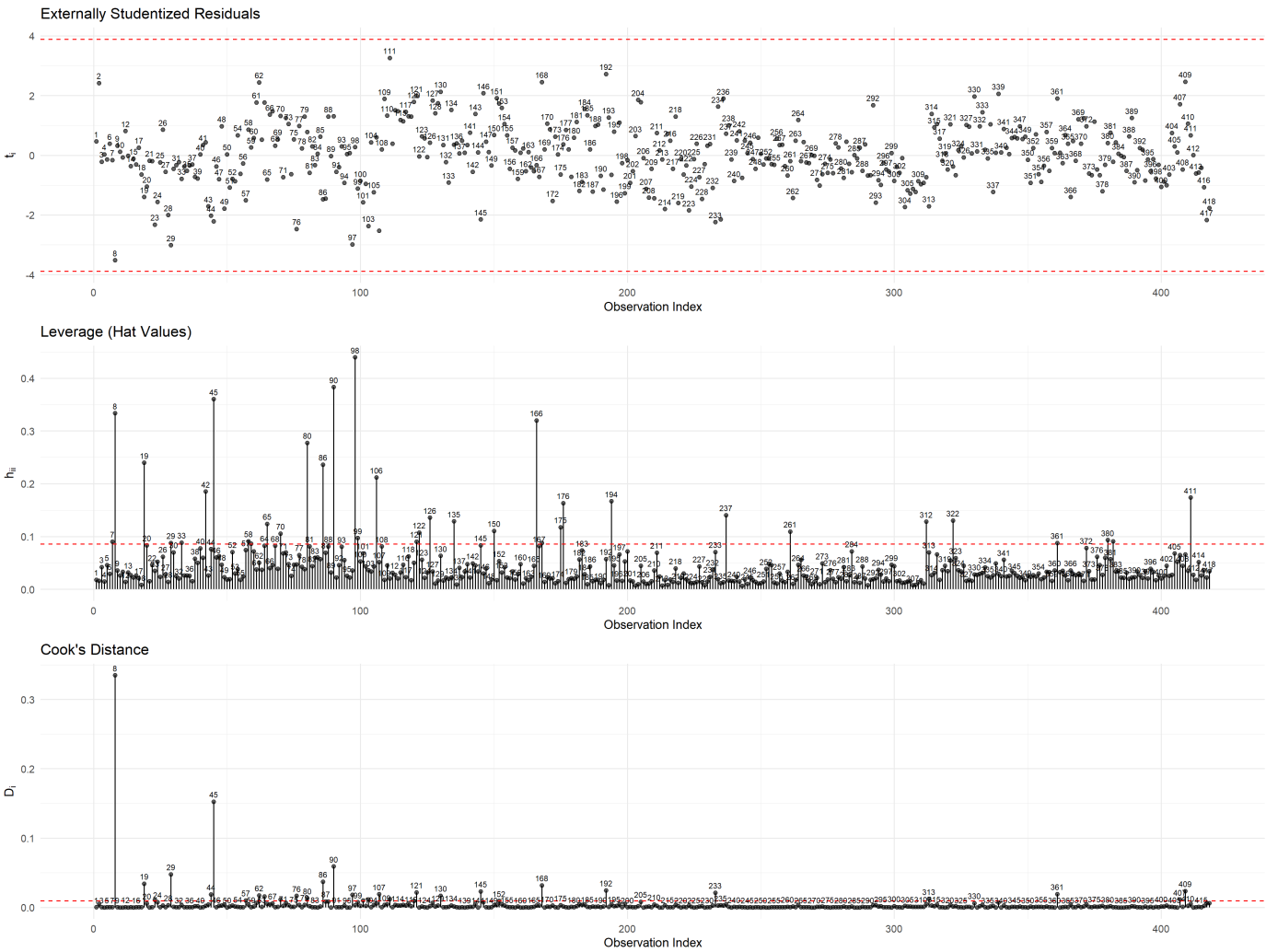


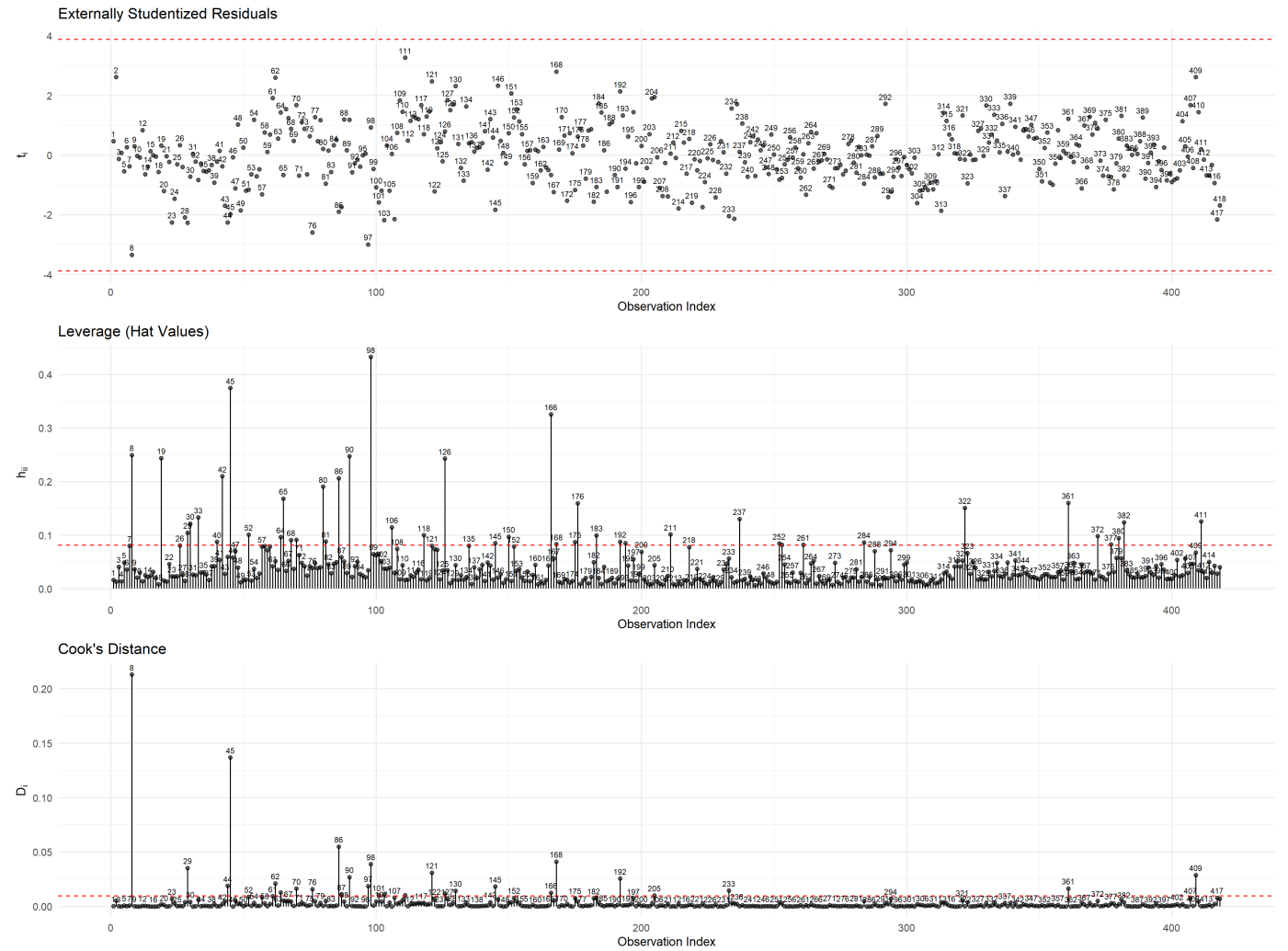


Here we have diagnostic plots for the two MLR models. I have chosen to eliminate the response vs predictor and residual vs predictor plots that were present for SLR diagnostics because we have many predictors here not just one. Also the residual vs response plot provides very similar information to those two plots.

First, looking at the residual vs $\log_price$ plot in both models, we can see that the confidence band very nicely follows the zero line with very little curve, indicating that both models are capturing a linear relationship. Now moving on to the absolute residual plot, the shape is similar for both candidates. There is a moderate curve for small fitted values and then the LOESS curve flattens out. In both models we would be unable to draw a horizontal line through the gray band, indicating that there is a small issue of non constant variance in both of these models. Looking at the histogram, in both models it is very much symmetric which is good. Finally for the QQ-plot we have very slightly heavy tails in both cases, but this is probably due to the heteroskedasticity we noticed earlier considering how symmetric the histograms are. Both models exhibit slight non constant variance, but perform very well in terms of linearity, normality, and non correlated errors. Visually there does not seem to be any particularly influential points but we will check that next.

As of right now heteroskedasticity seems to be the only issue, and to remedy that I would implement weighted least squares.





Both plots have the exact same result when we look for potentially influential points. There are no observations that have ESR outside of our expected boundary, but there are a few with high leverage, namely observation 98. As far as influential points, 8 has by far the highest Cook's distance, followed by 45 so these are the two most influential points. I should note that observation 8 is the point we flagged earlier as the only point that has $AC = 0$ and $Pool = 1$, so it is no surprise that it has been marked as influential.

coefficient	full_data	without_observation	change
(Intercept)	0.984	0.927	-0.057
Sqft	0.344	0.351	0.007
Year	0.346	0.383	0.037
Lot	0.171	0.176	0.005
Sqft:Bed	-0.084	-0.110	-0.026
Year:Quality2	-0.039	-0.050	-0.010
Year2	0.053	0.057	0.004
Pool	0.254	0.292	0.038
Bed2	0.015	0.041	0.026

coefficient	full_data	without_observation	change
Quality2	-1.082	-1.003	0.079
Quality3	-1.371	-1.224	0.147
Bath:Quality2	0.161	0.105	-0.055
Bath2	-0.050	-0.047	0.003
Year:AC	-0.127	-0.151	-0.024
Sqft:Quality2	0.199	0.194	-0.005
Bath	-0.020	0.023	0.044
Bed	0.033	0.042	0.009
AC	0.055	0.010	-0.045
coefficient	full_data	without_observation	change
(Intercept)	0.989	1.021	0.031
AC	0.249	0.203	-0.045
Pool	0.247	0.284	0.038
Year	0.168	0.166	-0.002
Lot	0.163	0.167	0.004
Bath2	-0.066	-0.061	0.005
Year2	0.046	0.047	0.002
Sqft:Year	0.083	0.093	0.010
Sqft:Quality2	0.354	0.390	0.036
Sqft:Quality3	0.423	0.539	0.116
Bath:Quality2	0.186	0.145	-0.041
Year:Quality3	0.139	0.179	0.040
AC:Quality2	-0.308	-0.266	0.042
Bath	-0.012	0.023	0.035
Quality2	-0.981	-1.015	-0.034
Quality3	-1.376	-1.260	0.116
Sqft	0.136	0.096	-0.039

Number of bathrooms at most influential point: 5

This is a sensitivity analysis after removing the most influential point. As we can see for the most part the changes are moderate, but there is one specific one to flag. The coefficient of Bath changed signs, which is definitely something to analyze further. As we see above the number of bathrooms at the most influential observation is 5. There are only 19 observations in this data set with that many bathrooms, which is a small percent, so getting rid of one of those observations is likely to create a significant sway in that coefficient. At this point we have had several issues with this specific observation so it would be wise to check if this is a

data entry error, but I have no way to do this. Despite the coefficient sign change we noticed, looking back at the diagnostic plots we don't seem to be having any major issue with this outlier, so we will just keep it.

Call:

```
lm(formula = log_price ~ Sqft + Year + Lot + `Sqft:Bed` + `Year:Quality2` +
    Year2 + Pool + Bed2 + Quality2 + Quality3 + `Bath:Quality2` +
    Bath2 + `Year:AC` + `Sqft:Quality2` + Bath + Bed + AC, data = data_train_full)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.0640	-0.2279	-0.0054	0.2190	1.1707

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	0.98404	0.12605	7.807	5.19e-14	***
Sqft	0.34414	0.05746	5.989	4.71e-09	***
Year	0.34562	0.06974	4.956	1.07e-06	***
Lot	0.17084	0.01939	8.811	< 2e-16	***
`Sqft:Bed`	-0.08352	0.02400	-3.480	0.000556	***
`Year:Quality2`	-0.03930	0.05268	-0.746	0.456091	
Year2	0.05317	0.01601	3.322	0.000976	***
Pool	0.25397	0.07196	3.529	0.000465	***
Bed2	0.01495	0.01592	0.940	0.347985	
Quality2	-1.08220	0.10336	-10.471	< 2e-16	***
Quality3	-1.37124	0.14856	-9.230	< 2e-16	***
`Bath:Quality2`	0.16081	0.06313	2.547	0.011231	*
Bath2	-0.04963	0.01425	-3.483	0.000551	***
`Year:AC`	-0.12680	0.06747	-1.880	0.060902	.
`Sqft:Quality2`	0.19880	0.06686	2.973	0.003123	**
Bath	-0.02027	0.04841	-0.419	0.675648	
Bed	0.03343	0.02493	1.341	0.180768	
AC	0.05520	0.07393	0.747	0.455672	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3669 on 400 degrees of freedom

Multiple R-squared: 0.8708, Adjusted R-squared: 0.8654

F-statistic: 158.6 on 17 and 400 DF, p-value: < 2.2e-16

Call:

```
lm(formula = log_price ~ AC + Pool + Year + Lot + Bath2 + Year2 +
    `Sqft:Year` + `Sqft:Quality2` + `Sqft:Quality3` + `Bath:Quality2` +
    `Year:Quality3` + `AC:Quality2` + Bath + Quality2 + Quality3 +
    Sqft, data = data_train_full)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.07114	-0.23237	-0.00235	0.23082	1.17421

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	0.98938	0.12738	7.767	6.78e-14	***
AC	0.24870	0.07211	3.449	0.000623	***
Pool	0.24655	0.07177	3.435	0.000654	***
Year	0.16814	0.03101	5.423	1.02e-07	***
Lot	0.16320	0.01965	8.304	1.57e-15	***
Bath2	-0.06600	0.01371	-4.813	2.11e-06	***
Year2	0.04555	0.01540	2.958	0.003282	**
`Sqft:Year`	0.08266	0.02841	2.909	0.003823	**
`Sqft:Quality2`	0.35444	0.07781	4.555	6.95e-06	***
`Sqft:Quality3`	0.42312	0.11332	3.734	0.000216	***
`Bath:Quality2`	0.18565	0.06296	2.949	0.003379	**
`Year:Quality3`	0.13913	0.07279	1.912	0.056649	.
`AC:Quality2`	-0.30809	0.12135	-2.539	0.011500	*
Bath	-0.01221	0.04705	-0.259	0.795447	
Quality2	-0.98106	0.15688	-6.254	1.03e-09	***
Quality3	-1.37604	0.13379	-10.285	< 2e-16	***
Sqft	0.13578	0.06852	1.981	0.048219	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3671 on 401 degrees of freedom

Multiple R-squared: 0.8704, Adjusted R-squared: 0.8652

F-statistic: 168.3 on 16 and 401 DF, p-value: < 2.2e-16

Here are both of the model summaries and we will analyze these for evidence of multicollinearity. In the forward selection model, 11 out of 17 of our predictors have significant t-tests and the F-test is significant with a large F statistic which is a good sign. Additionally none of the standard errors are alarming for any coefficient. The fact that we have several predictors with an insignificant t-value could be a sign of multicollinearity. The backward selection model is much better with 14 out of 16 predictors being significant. The F-test is again significant and the F statistic is higher in this case. The standard errors are once again very reasonable, so there are no indicators of multicollinearity in this model.

Sqft	Year	Lot	`Sqft:Bed`	`Year:Quality2`
10.226780	15.062558	1.164384	2.759777	3.437865
Year2	Pool	Bed2	Quality2	Quality3
2.420350	1.103935	2.133826	8.162530	14.555357
`Bath:Quality2`	Bath2	`Year:AC`	`Sqft:Quality2`	Bath
4.076103	1.532128	9.534480	4.651492	7.257388
Bed	AC			
1.925052	2.311011			
AC	Pool	Year	Lot	Bath2
2.197317	1.097126	2.975238	1.195326	1.417475
Year2	`Sqft:Year`	`Sqft:Quality2`	`Sqft:Quality3`	`Bath:Quality2`
2.238618	2.666452	6.294970	7.947672	4.051156
`Year:Quality3`	`AC:Quality2`	Bath	Quality2	Quality3
5.350159	11.394191	6.849970	18.791045	11.796760

Sqft
14.530541

Looking at the VIFs there is clearly multicollinearity in both models. There are several predictors Sqft, Year, Quality2, and Quality3 have large VIFs across both models as well as several of their interaction terms. This is not particularly surprising, as we would expect correlation between terms and their interactions so this is an expected consequence of fitting the more complex model structure. The highest VIF is Quality2 in the backward selection model, which is also expected. Earlier we discussed how Quality2 is very strongly related with several other predictors in the pool, so it comes at no surprise it has a high VIF. Given the expected and structural nature of these high VIF quantities, I am going to choose to do weighted least squares to address non constant variance. While ridge regression would help with the correlation that is present in this model, I think I would get less of a benefit from doing this since multicollinearity is a structural tradeoff that I am accepting by fitting a model with many quadratic and interaction terms.

Forward Selection: OLS vs WLS

method	term	Estimate	Std. Error	t value	Pr(> t)
OLS	(Intercept)	0.9840	0.1260	7.8068	0.0000
OLS	Sqft	0.3441	0.0575	5.9888	0.0000
OLS	Year	0.3456	0.0697	4.9559	0.0000
OLS	Lot	0.1708	0.0194	8.8106	0.0000
OLS	Sqft:Bed	-0.0835	0.0240	-3.4801	0.0006
OLS	Year:Quality2	-0.0393	0.0527	-0.7460	0.4561
OLS	Year2	0.0532	0.0160	3.3219	0.0010
OLS	Pool	0.2540	0.0720	3.5292	0.0005
OLS	Bed2	0.0150	0.0159	0.9396	0.3480
OLS	Quality2	-1.0822	0.1034	-10.4706	0.0000
OLS	Quality3	-1.3712	0.1486	-9.2304	0.0000
OLS	Bath:Quality2	0.1608	0.0631	2.5473	0.0112
OLS	Bath2	-0.0496	0.0143	-3.4828	0.0006
OLS	Year:AC	-0.1268	0.0675	-1.8795	0.0609
OLS	Sqft:Quality2	0.1988	0.0669	2.9735	0.0031
OLS	Bath	-0.0203	0.0484	-0.4187	0.6756
OLS	Bed	0.0334	0.0249	1.3407	0.1808
OLS	AC	0.0552	0.0739	0.7467	0.4557
WLS	(Intercept)	0.9898	0.1275	7.7619	0.0000
WLS	Sqft	0.3508	0.0582	6.0278	0.0000
WLS	Year	0.3385	0.0679	4.9837	0.0000
WLS	Lot	0.1670	0.0197	8.4710	0.0000

method	term	Estimate	Std. Error	t value	Pr(> t)
WLS	Sqft:Bed	-0.0820	0.0246	-3.3318	0.0009
WLS	Year:Quality2	-0.0471	0.0524	-0.8984	0.3695
WLS	Year2	0.0476	0.0161	2.9583	0.0033
WLS	Pool	0.2528	0.0743	3.4014	0.0007
WLS	Bed2	0.0155	0.0160	0.9663	0.3345
WLS	Quality2	-1.0694	0.1082	-9.8882	0.0000
WLS	Quality3	-1.3474	0.1522	-8.8535	0.0000
WLS	Bath:Quality2	0.1480	0.0621	2.3811	0.0177
WLS	Bath2	-0.0514	0.0144	-3.5720	0.0004
WLS	Year:AC	-0.1288	0.0660	-1.9514	0.0517
WLS	Sqft:Quality2	0.1928	0.0682	2.8290	0.0049
WLS	Bath	-0.0035	0.0452	-0.0771	0.9385
WLS	Bed	0.0267	0.0241	1.1085	0.2683
WLS	AC	0.0443	0.0686	0.6453	0.5191

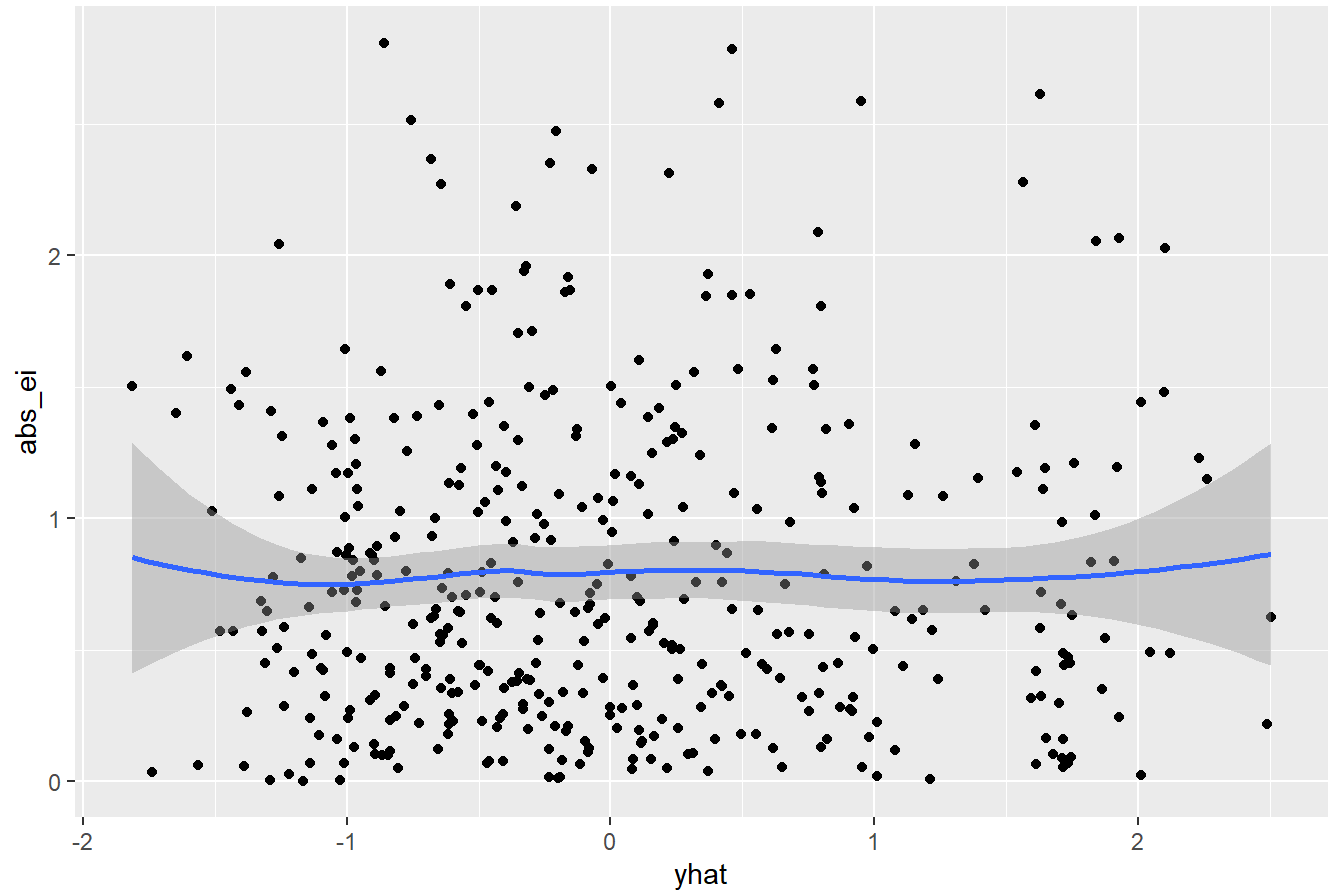
Backward Selection: OLS vs WLS

method	term	Estimate	Std. Error	t value	Pr(> t)
OLS	(Intercept)	0.9894	0.1274	7.7671	0.0000
OLS	AC	0.2487	0.0721	3.4488	0.0006
OLS	Pool	0.2466	0.0718	3.4353	0.0007
OLS	Year	0.1681	0.0310	5.4227	0.0000
OLS	Lot	0.1632	0.0197	8.3037	0.0000
OLS	Bath2	-0.0660	0.0137	-4.8134	0.0000
OLS	Year2	0.0455	0.0154	2.9578	0.0033
OLS	Sqft:Year	0.0827	0.0284	2.9094	0.0038
OLS	Sqft:Quality2	0.3544	0.0778	4.5553	0.0000
OLS	Sqft:Quality3	0.4231	0.1133	3.7339	0.0002
OLS	Bath:Quality2	0.1856	0.0630	2.9486	0.0034
OLS	Year:Quality3	0.1391	0.0728	1.9115	0.0566
OLS	AC:Quality2	-0.3081	0.1214	-2.5388	0.0115
OLS	Bath	-0.0122	0.0470	-0.2594	0.7954
OLS	Quality2	-0.9811	0.1569	-6.2535	0.0000
OLS	Quality3	-1.3760	0.1338	-10.2848	0.0000
OLS	Sqft	0.1358	0.0685	1.9815	0.0482

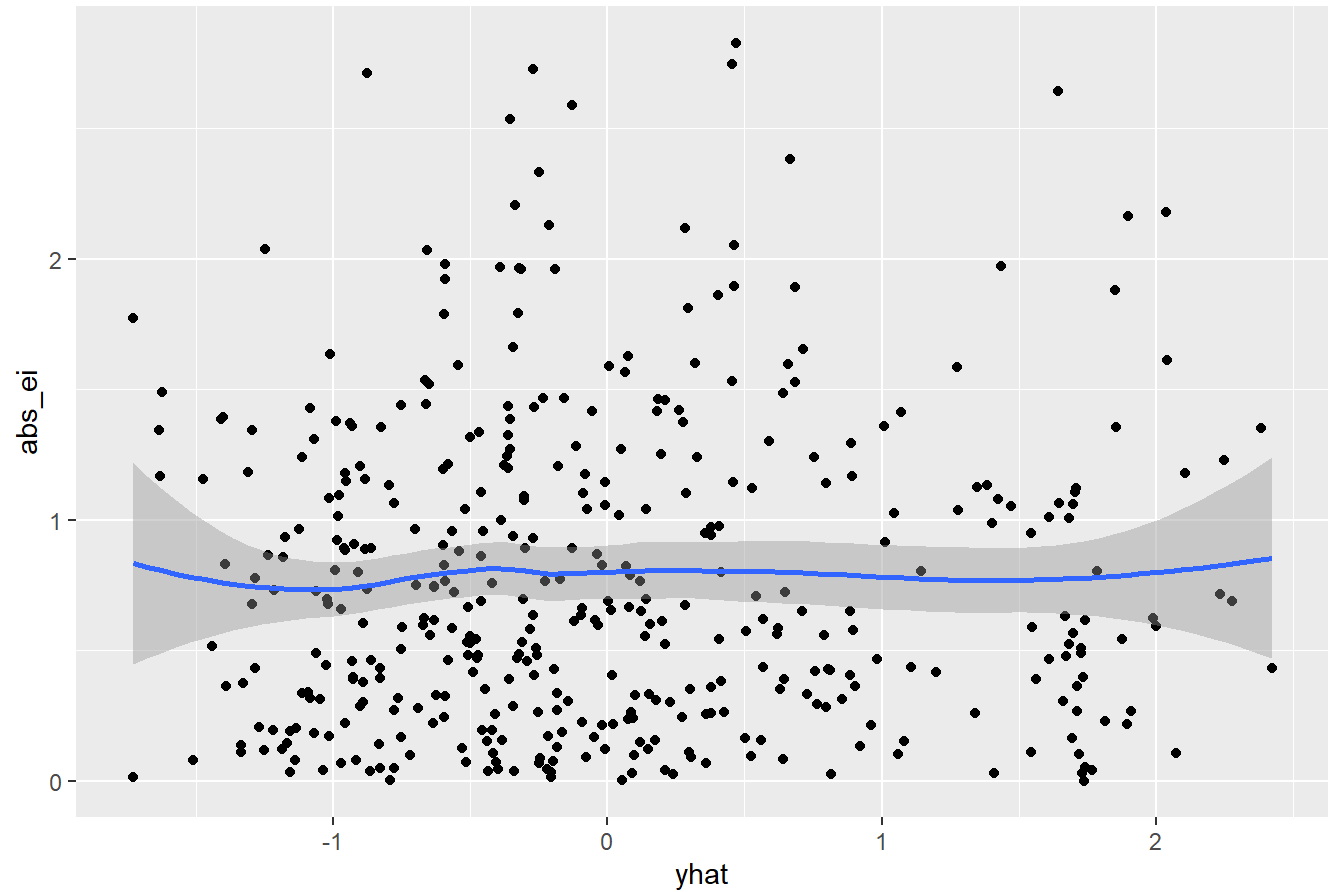
method	term	Estimate	Std. Error	t value	Pr(> t)
WLS	(Intercept)	1.0377	0.1308	7.9361	0.0000
WLS	AC	0.2115	0.0612	3.4538	0.0006
WLS	Pool	0.2474	0.0740	3.3432	0.0009
WLS	Year	0.1624	0.0305	5.3185	0.0000
WLS	Lot	0.1600	0.0200	8.0068	0.0000
WLS	Bath2	-0.0665	0.0140	-4.7366	0.0000
WLS	Year2	0.0391	0.0149	2.6194	0.0091
WLS	Sqft:Year	0.0792	0.0285	2.7741	0.0058
WLS	Sqft:Quality2	0.3618	0.0842	4.2973	0.0000
WLS	Sqft:Quality3	0.4298	0.1153	3.7276	0.0002
WLS	Bath:Quality2	0.1745	0.0621	2.8087	0.0052
WLS	Year:Quality3	0.1466	0.0685	2.1389	0.0330
WLS	AC:Quality2	-0.2613	0.1165	-2.2435	0.0254
WLS	Bath	-0.0027	0.0447	-0.0595	0.9526
WLS	Quality2	-1.0352	0.1610	-6.4292	0.0000
WLS	Quality3	-1.3694	0.1409	-9.7219	0.0000
WLS	Sqft	0.1364	0.0751	1.8147	0.0703

For both forward and backwards, the coefficients are stable from OLS to WLS, showing very minimal change across the board, but there are a few things to point out here. In the forward model, none of the coefficients flip, while in the backward model Sqft goes from barely significant to insignificant and Year:Quality3 goes from insignificant to significant. The standard errors change in a nonuniform way, which is exactly what we expect from WLS. The new weights are determined by the spread of each observation, so the change will be variable. The fact that Sqft is not significant in the backward model is interesting because we picked it out as our most correlated predictor in our original diagnostics. This could be because it is strongly correlated with other terms, so the interactions explain away what it could have explained on its own.

Weighted |Residuals| vs Fitted - Forward Selection WLS Model



Weighted |Residuals| vs Fitted - Backward Selection WLS Model



After the getting the least squares weights we can see that we now have beautiful constant variance across all of the data which is the exact fix that we wanted from this remedial measure.

In-sample and out-of-sample model comparison

Model	Predictors	Adj_R2	AIC	BIC	PRESS_RMSE	Test_RMSE
Forward (OLS)	17	0.8654	367.6963	444.3704	0.3817	0.3954
Forward (WLS)	17	0.8624	341.2021	417.8763	0.3796	0.3946
Backward (OLS)	16	0.8652	367.0728	439.7114	0.3793	0.4130
Backward (WLS)	16	0.8604	341.6069	414.2456	0.3779	0.4106

10-fold cross-validation (training data)

Model	CV_RMSE	CV_RMSE_SD	CV_MAE
Forward (OLS)	0.3841	0.0342	0.2983
Backward (OLS)	0.3811	0.0299	0.2965

As you can see all four of our final models are very similar on this test data. However, I am going to pick the forward WLS model as the best overall. First of all since it is a WLS model it does not violate the heteroskedasticity assumption and this model had the lowest test RMSE overall. Between all four models the adjusted R^2 are very similar with negligible difference. Looking at BIC and AIC, the forward WLS model

has the lowest AIC and second lowest BIC, but the difference is again very small. For PRESS RMSE the story is the same this model has the second lowest but the difference to the lowest is insignificant.

Cross validation was only done on the OLS models because the WLS models used the full training set to develop weights, so trying to do cross validation would have some data leakage and get faulty results. As we can see the backward model had better results but the difference is once again very small, so even though backward is better here I will still take the forward WLS model overall. As far as interpretability, the models only differ by one predictor, so we don't really lose much interpretation between the two models, and as such I am still confident in selection the larger model.

Call:

```
lm(formula = formula(forward_fit), data = data_train_full, weights = wls_weight_fwd)
```

Weighted Residuals:

Min	1Q	Median	3Q	Max
-3.5424	-0.8161	-0.0056	0.7384	3.5094

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	0.989796	0.127520	7.762	7.06e-14	***
Sqft	0.350793	0.058196	6.028	3.78e-09	***
Year	0.338454	0.067912	4.984	9.30e-07	***
Lot	0.166999	0.019714	8.471	4.70e-16	***
`Sqft:Bed`	-0.081954	0.024598	-3.332	0.000943	***
`Year:Quality2`	-0.047067	0.052391	-0.898	0.369524	
Year2	0.047633	0.016102	2.958	0.003277	**
Pool	0.252827	0.074331	3.401	0.000738	***
Bed2	0.015459	0.015999	0.966	0.334481	
Quality2	-1.069447	0.108154	-9.888	< 2e-16	***
Quality3	-1.347446	0.152194	-8.853	< 2e-16	***
`Bath:Quality2`	0.147952	0.062135	2.381	0.017727	*
Bath2	-0.051360	0.014379	-3.572	0.000397	***
`Year:AC`	-0.128783	0.065996	-1.951	0.051709	.
`Sqft:Quality2`	0.192845	0.068167	2.829	0.004904	**
Bath	-0.003488	0.045214	-0.077	0.938546	
Bed	0.026685	0.024074	1.108	0.268327	
AC	0.044282	0.068622	0.645	0.519102	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.261 on 400 degrees of freedom

Multiple R-squared: 0.868, Adjusted R-squared: 0.8624

F-statistic: 154.8 on 17 and 400 DF, p-value: < 2.2e-16

$$\begin{aligned}\log(\text{Price}_i) = & \beta_0 + \beta_1 \text{Sqft}_i + \beta_2 \text{Year}_i + \beta_3 \text{Lot}_i + \beta_4 (\text{Sqft} \times \text{Bed})_i \\ & + \beta_5 (\text{Year} \times \text{Quality}_2)_i + \beta_6 \text{Year}_i^2 + \beta_7 \text{Pool}_i + \beta_8 \text{Bed}_i^2 \\ & + \beta_9 \text{Quality}_{2,i} + \beta_{10} \text{Quality}_{3,i} + \beta_{11} (\text{Bath} \times \text{Quality}_2)_i \\ & + \beta_{12} \text{Bath}_i^2 + \beta_{13} (\text{Year} \times \text{AC})_i + \beta_{14} (\text{Sqft} \times \text{Quality}_2)_i \\ & + \beta_{15} \text{Bath}_i + \beta_{16} \text{Bed}_i + \beta_{17} \text{AC}_i + \varepsilon_i.\end{aligned}$$

Above is the final model selection which is a forward selection model that we then used weighted least squares on to fix issues with heteroskedasticity. This model contains 17 predictors including quadratic and interaction terms. The fitted model indicates that a residence's sale price is influenced by a nontrivial combination of characteristics such as what features are present in the residence as well as basic statistics. The inclusion of quadratic terms for Year, Bed, and Bath indicates that the relationships between these characteristics and the price is not purely linear. The presence of several interaction terms also indicates that the effect of one predictor may depend on the value of another. For example in the Sqft and Bed interaction, the coefficient is negative. This indicates that the increase in sales price due to square footage actually decreases as the number of bedrooms increases. The fact that we have significant coefficients for Sqft:Quality₂, Year:Quality₂, and Bath:Quality₂ indicates that the increase in sale price based on these predictors does differ across different levels of residence quality. Overall this indicates that in many cases the structural characteristics of the residence cannot be interpreted independently of home quality.

The final model has an R² value of 0.8624, indicating that roughly 86 percent of the variability in the log of price is due to the selected characteristics in the model. This model also achieved a test RMSE of 0.3946 which was almost identical to the cross-validation error that was observed using the OLS version of this model. This is an indicator that our model is generalizing very well and is not over fitting the data that it was trained on.

There are two main issues with this model. One was the observed multicollinearity in the fit of the OLS model, which would still be present in the WLS model. We observed high VIF values above 10 for several of the predictors. This was expected considering the amount of interaction and quadratic terms in the model, as those would naturally be correlated with their corresponding main effects. However this is still an issue, and in the future I would consider adding a ridge penalty to mitigate this issue. The second issue is that this model is complex and not trivial to interpret. While the coefficients individually tell a clear story, there are 17 of them, so it is difficult to understand the full picture. Another thing to test in the future would be comparative analysis of this model against less complex models to see if there is a fit that is more interpretable with similar predictive power and stability.

As far as the research question, it is clear that a combination of structural features and selected interactions between them provide an accurate prediction of residential sale price. There is no single characteristic that clearly determines the value, and we have to investigate each coefficient to understand how the interactions between the most important characteristics determine what the residence will eventually be sold for. Overall we can see that the most important predictors are Sqft, Year, Lot, Bed, Pool, Bath, AC, and Quality. These predictors provide accurate insight into what the home will be worth when we consider them in isolation as well as selected interactions between each other.